Overparameterization and Efficient Adversarial Training of Neural Networks

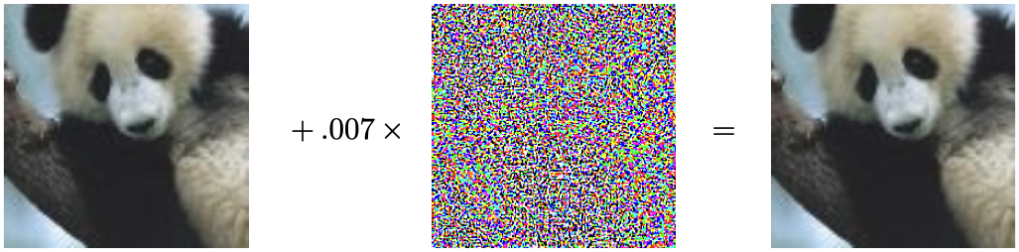
Krishna Acharya
Georgia Institute of Technology
krishna.acharya@gatech.edu

Alexander Bukharin
Georgia Institute of Technology
abukharin3@gatech.edu

Tyler LaBonte
Georgia Institute of Technology
tlabonte@gatech.edu

1 Introduction

Over the past decade deep neural networks have achieved remarkable success in reinforcement learning, computer vision, and natural language processing tasks. Due to this success, deep neural networks have been used in a number of safety-critical applications such as autonomous driving [Grigorescu et al., 2020], medical imaging [Suzuki, 2017], and speech recognition [Deng et al., 2013]. Although the deep neural networks used in these applications can accurately classify inputs from the training dataset, it has been shown that they cannot classify adversarially perturbed inputs [Szegedy et al., 2013]. More specifically, given an input that the neural network accurately classifies, it has been shown that we can add a small (i.e., imperceptible to humans) perturbation to the input such that the neural network will incorrectly classify the perturbed input. An example of an adversarial attack can be seen in Figure 1. This phenomenon poses a serious safety concern for the deployment of deep neural networks, as malicious inputs may be used to manipulate the network's behavior.



$\boldsymbol{x}$
"panda"
57.7% confidence

$\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"nematode"
8.2% confidence

$\boldsymbol{x} + \epsilon\text{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"gibbon"
99.3 % confidence

Figure 1: Example of an adversarial attack from Goodfellow et al. [2014]. The model used is GoogLeNet [Szegedy et al., 2015].

In light of the vulnerabilities of deep neural networks to adversarial perturbations, many methods have been developed to train adversarially robust deep neural networks. In this report we will discuss how robust neural network training can be formulated as a minimax optimization problem, the computational cost of solving the minimax problem, and recent approaches for efficient adversarial training. In addition, we will discuss how the Lipschitz constant of a function affects its robustness.

36th Conference on Neural Information Processing Systems (NeurIPS 2022).

2 Adversarial Training

2.1 Problem Formulation

Typically, deep neural networks are trained via empirical risk minimization (ERM) over a dataset $\mathcal{D} = (\mathcal{X}, \mathcal{Y})$, where $\mathcal{X} = \{x_1, ..., x_N\}$ and $\mathcal{Y} = \{y_1, ..., y_N\}$. Given a neural network $f_\theta(\cdot)$ and a loss function $\mathcal{L}_\theta(x, y) = \sum_{i=1}^{N} \ell(f_\theta(x), y)$, the model $f_\theta(\cdot)$ is trained by minimizing $\mathcal{L}_\theta(x, y)$.

If the dataset is large enough, this standard training procedure will result in neural networks that can accurately classify instances from the training set and generalize to some new instances. However, neural networks produced by this procedure have also been found to be sensitive to small perturbations to the input. In particular, for some instances (x, y) in the dataset, we can generate a small perturbation δ such that $D(f_\theta(x), f_\theta(x + \delta))$ is large for some distance metric D. This is problematic, because for most real world applications the optimal classifier f^* is relatively smooth, and its output should not change drastically from a small change to its input. In applications, this means neural networks can be manipulated by only changing their input slightly.

Based on the shortcomings of ERM, Madry et al. [2017] propose minimizing the adversarial risk

$$\min_{\theta} \max_{\delta_i : \|\delta_i\| \leq \epsilon} \sum_{i=1}^{N} \ell(f_\theta(x_i + \delta_i), y_i). \tag{1}$$

The inner maximization problem of (1) tries to find the perturbation δ_i that will result in the largest loss for $f_\theta(\cdot)$ on the pair (x_i, y_i). The outer maximization problem then tries to learn θ that accurately classifies the adversarial examples. Note that when ϵ is set to 0, the minimax optimization problem reduces to ERM. Therefore, solving (1) will result in a model that accurately classify the clean training data (x_i, y_i) and is robust to training set perturbations with norm less than ϵ.

2.2 Minimax Optimization

Even though solving (1) will result in an accurate and robust model, (1) is a non-convex non-concave minimax problem and is therefore NP-hard. To try and solve (1), we may use Danskin's theorem, a fundamental result from optimization. In essence, Danskin's theorem states that

$$\begin{aligned} &\nabla_\theta \max_{\delta \in \Delta} \mathcal{L}_\theta(x + \delta, y; \theta) = \nabla_\theta \mathcal{L}_\theta(x + \delta^\star, y; \theta) \\ &\text{where } \delta^\star = \max_{\delta \in \Delta} \mathcal{L}_\theta(x + \delta, y; \theta). \end{aligned}$$

Thus we can optimize through the max of Eq (1), but only by using an exact maximizer $\delta^\star$. Although there is no guarantee that such a maximizer can be computed, it has been shown empirically that projected gradient descent consistently finds a powerful maximizer δ [Madry et al., 2017]. This phenomenon may be thought of from the perspective of well concentrated loss values [Madry et al., 2017]. Just as neural networks can be trained by first order methods because neural network local minima have similar loss values, powerful attacks may be generated by PGD because neural network local maxima also have similar loss values.

3 Efficient Adversarial Training

There are two main paradigms for the inner maximization which the adversarial training community has converged to:

- **Fast Gradient Sign Method (FGSM):** Finds the steepest direction which maximizes the inner objective (its first-order Taylor approximation) and takes one-step along that direction [Goodfellow et al., 2014]. The main benefit of FGSM is that it is relatively efficient, as only one step of gradient ascent is used for the inner objective. Although training with FGSM can improve robustness, taking more gradient steps can lead to more powerful attacks [Tramèr et al., 2017, Madry et al., 2017]. Therefore training with FGSM may not guarantee robustness.

- **Multi-step Projected Gradient Descent (PGD):** Repeated projected gradient descent [Madry et al., 2017]. PGD has been found to generate powerful attacks, and training with PGD can effectively defend against PGD and FGSM attacks [Madry et al., 2017]. PGD is also empirically shown to have stable training (i.e not requiring a lot of hyperparameter tuning) and robust to many attacks during testing [Rice et al., 2020]. However, PGD typically requires 10 to 50 gradient steps to solve the inner maximization problem [Madry et al., 2017, Sriramanan et al., 2021]. This means that PGD training of a neural network will take 10 times to 50 times longer than standard training via SGD. Such a computationally expensive training procedure is not feasible for many modern neural networks, which contain billions of parameters and are trained on huge datasets.

Due to the high computational cost of PGD and the relative weakness of FGSM, there is a need for methods that can tradeoff efficiency and robustness, two works address these issues:

- **Principled one-step:** For FGSM adversarial training Andriushchenko and Flammarion [2020] prevent catastrophic overfitting by explicitly maximizing the gradient alignment inside the perturbation set. This improves the quality of their FGSM solution and they have interesting theory motivating their idea.
- **Principled multi-step:** Wang et al. [2019], is a type of curriculum learning, which gives easier adversarial examples first i.e run PGD with lesser iterations initially then ramp up the number of iterations. Compared to other dynamic training papers they do the above in a more principled way. The potential function for the inner maximization is a first order stationary condition (FOSC). This criteria is agnostic when compared to training loss (when we change model-sizes, PGD steps) and we will use it as a rough metric to measure convergence quality.

4 First-Order Stationary Condition as a convergence criterion

Wang et al. [2019] propose the following First-Order Stationary Condition (FOSC) as the convergence criterion for the inner maximization problem:

$$c(\mathbf{x}^k) = \max_{\mathbf{x}\in\mathcal{X}} \langle \mathbf{x} - \mathbf{x}^k, \nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle, \tag{2}$$

Here $\mathbf{x}^0$ is the starting natural image(which is progressively pertubed to an adversarial example), $\mathbf{x}^k$, is the kth iterate of any algorithm doing the inner maximization(for e.g. k^{th} iterate of PGD). $\mathcal{X} = \{\mathbf{x} | \|\mathbf{x} - \mathbf{x}^0\|_\infty \leq \epsilon\}$ is the input domain of the ϵ-ball around starting natural image $\mathbf{x}^0$, $f(\boldsymbol{\theta}, \mathbf{x}^k) = \ell(h_{\boldsymbol{\theta}}(\mathbf{x}^k), y)$. Note that $c(\mathbf{x}^k) \geq 0$ [1].

The criterion FOSC in Eq. (2) can be shown to have the following closed-form solution:

$$\begin{aligned}
c(\mathbf{x}^k) &= \max_{\mathbf{x}\in\mathcal{X}} \langle \mathbf{x} - \mathbf{x}^k, \nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle \\
&= \max_{\mathbf{x}\in\mathcal{X}} \langle \mathbf{x} - \mathbf{x}^0 + \mathbf{x}^0 - \mathbf{x}^k, \nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle \\
&= \max_{\mathbf{x}\in\mathcal{X}} \langle \mathbf{x} - \mathbf{x}^0, \nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle \\
&\qquad + \langle \mathbf{x}^k - \mathbf{x}^0, -\nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle \\
&= \epsilon \|\nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\|_1 - \langle \mathbf{x}^k - \mathbf{x}^0, \nabla_{\mathbf{x}} f(\boldsymbol{\theta}, \mathbf{x}^k)\rangle.
\end{aligned}$$

A smaller value of $c(\mathbf{x}^k)$ indicates a better solution of the inner maximization (or equivalently, better convergence quality of the adversarial example $\mathbf{x}^k$).
$c(\mathbf{x}^k) = 0$ can be achieved iff

1. $\nabla f(\boldsymbol{\theta}, \mathbf{x}^k) = 0$, *i.e.*, $\mathbf{x}^k$ is a stationary point in the interior of $\mathcal{X}$
2. $\mathbf{x}^k - \mathbf{x}^0 = \epsilon \cdot \text{sign}(\nabla f(\boldsymbol{\theta}, \mathbf{x}^k))$, that is, local maximum point of $f(\boldsymbol{\theta}, \mathbf{x}^k)$ is reached on the boundary of $\mathcal{X}$

Wang et al. [2019] Argue via many empirical plots that this convergence criteria is the one to focus on when doing the inner maximization. Earlier works monitored the loss sequences, and these can converge to different values, whereas we know convergence criteria can only converge to 0

[1] Because $\mathbf{x}^k$ is always in $\mathcal{X}$, thus $x = \mathbf{x}^k$ is a feasible point, and the max can be only greater

5 Lipschitz Continuity and Robustness

Another interesting direction is how overparameterization of massive neural networks affects adversarial robustness and adversarial training. Utilizing these properties may give insight in how to design more efficient adversarial robustness algorithms with provable guarantees – in essence, we may be able to trade off some computational complexity in adversarial training for the increased memory and expressive complexity of larger neural networks.

Research in this area takes a functional perspective of neural networks and commonly studies robustness in the sense of the Lipschitz constant of the function learnt by the model. Models with smaller Lipschitz constant change more gradually within an ϵ-ball around the input, so they are presumably more resistant to adversarial attacks.

An exciting new result in this area is Bubeck and Sellke's universal law of robustness Bubeck and Sellke [2021]. They prove that there is a tradeoff between robustness and overparameterization in the following sense: To smoothly fit n datapoints in d dimensions, one either has a small model (with n parameters) which must then be non-robust, or a large model (with nd parameters) which is constant Lipschitz. In this sense, overparameterization is *necessary* for robustness.

This property is relevant to our project in efficiency of adversarial training because overparameterized neural networks are both (1) theoretically capable of Lipschitz robustness, and (2) have useful optimization properties which may enable more efficient adversarial training algorithms. For an example of the latter, Bubeck et al. [2021] prove that subexponential width random two-layer ReLU neural networks can find adversarial examples in a *single step* of PGD with high probability. This provides some theoretical guarantees to the FGSM algorithm mentioned previously. It is an interesting question whether we can develop an FGSM variant which utilizes the smoothness of overparameterized models to succeed in solving the inner maximization in a single gradient step.

6 Overparametrization and Adversarial Robustness

In this section we will review work related to adversarial robustness and overparametrization. In particular, we examine three papers that are most related to our proposed experiments.

6.1 Convergence of Adversarial Training in Overparametrized Neural Networks

This paper studies the convergence of adversarial training in overparametrized neural networks [Gao et al., 2019]. Specifically, this paper provides some amazing theoretical contributions to the study of adversarial training and overparametrization. Using the recent connection to Neural Tangent Kernels (NTK, [Jacot et al., 2018]), Gao et al. [2019] show the following:

- For a H-layer deep ReLU network and any attack algorithm $\mathcal{A}$, if the width of the network m is large enough, then projected gradient descent converges to a network that can achieve robust loss within ϵ of the optimal robust loss. In particular, the requirement on width is polynomial in the depth H and input dimension. In addition, they show that the projection step can be dropped in the case of two layer neural networks.
- The VC-dimension of models that can robustly interpolate n points is lower bounded by $\Omega(nd)$, where d is the data dimension. In contrast, some neural networks can interpolate n points with only $O(n)$ parameters, and have VC-dimension $O(n\log n)$. This implies that robustly interpolating data requires more parameters.

Although this provides many insights into over-parameterization and robustness, our work differs from it in several ways. First, our study will be largely empirical, as opposed to Gao et al. [2019] which only provides theory. In addition, Gao et al. [2019] do not study the efficiency of adversarial training, how model size effects robustness (they only study the minimum number of parameters needed for robust interpolations), or the FOSC criteria.

6.2 The curse of overparametrization in adversarial training: Precise analysis of robust generalization for random features regression

Hassani and Javanmard [2022b] studies how overparametrization effects adversarial robustness in two-layer random feature neural networks. Random feature networks are networks where the first layer is frozen, and learning only occurs in the second layer. They find that in such networks, overparametrization can actually hurt robust performance. This implies that increasing network size can actually increase the vulnerability to adversarial attacks. However, we find that the setting of two-layer random feature networks is extremely limiting. Therefore, we will study if the findings of this paper hold true on real world networks used in applications.

6.3 Do Wider Neural Networks Really Help Adversarial Robustness?

This work is most similar to ours. Wu et al. [2021] empirically study the robust accuracy of different width adversarially-trained neural networks. From this study, they find that the robust loss can actually increase for wider neural networks, although natural loss decreases. In addition, they theoretically show that the input gradient norm $\|\nabla \mathcal{L}(f(x'), y)\| = O(\sqrt{mH})$, where m is network width and H is the network depth. This implies that the local Lipschitz constant of a neural network is related to the number of parameters in the network.

Different from our study, their empirical study focuses on adversarially trained models. However, most models used today are not adversarially trained due to the large cost of minimax optimization. Therefore we will study if the finding of Wu et al. [2021] hold for industry standard networks, both with and without adversarial training. In addition, we will investigate how model size effects the efficiency of adversarial training.

7 Our Approach

We propose a joint theoretical and empirical investigation of the efficiency of adversarial training in overparameterized deep neural networks. In particular, we plan to extend and unify the observations of Wang et al. [2019], Bubeck et al. [2021], Bubeck and Sellke [2021] by performing experiments, then do theoretical analysis to understand our results or develop new algorithms. The key questions we would like to answer are:

1. **Does overparameterization help or hurt adversarial robustness?** Recent research [Bubeck and Sellke, 2021] suggests that overparameterization is not only useful, but necessary for achieving adversarial robustness in the sense of a constant Lipschitz factor. On the other hand, Hassani and Javanmard [2022a] show via a theoretical analysis that overparameterization can provably harm robust generalization in random features regression models. Due to this conflicting evidence, there is a need for a large-scale empirical study and baselines similar to what Sagawa et al. [2020] did for the related question of group robustness under spurious correlations (indeed, they found that overparameterization is harmful in many practical scenarios).
2. **Does overparameterization increase the efficiency of adversarial training?** Bubeck et al. [2021] show that for a certain class of overparameterized models, PGD can provably find an optimal adversarial example in a single step with high probability. However, there is a gap to implementation, as the results of Bubeck et al. [2021] held for random-features two-layer networks whose behavior may not correlate with deep models trained by SGD. Note that this is a separate question from (1), as it is an interesting possibility that overparameterized models have the theoretical capability to achieve adversarial robustness, but it may be computationally infeasible.
3. **Does the FOSC criteria properly capture the behavior of adversarial training in the overparameterized regime?** Wang et al. [2019] propose the first-order stationarity condition (FOSC) for measuring the optimality of an adversarial example, but their experiments do not show its correlation with overparameterization. If this relationship is not captured, then it may suggest that FOSC is lacking in its ability to predict the efficacy of an adversarial perturbation, and hence is a suboptimal stopping criteria for PGD. However, if larger models can achieve a better FOSC value in the same amount of steps, it evidences a promising

direction for developing algorithms which trade-off between computational complexity of adversarial training and expressive complexity of overparameterized networks.

To answer these three key questions, we will implement the following experiments. Our main dataset for evaluation will be CIFAR-10 [Krizhevsky, 2009], a dataset of 60,000 32×32 color images in 10 classes which is broadly used as a benchmark dataset in computer vision.

1. **Models.** We will test two different families of models, time and compute permitting. The first family of models is 5-layer CNNs with doubling widths, a commonly-used model class in works such as Nakkiran et al. [2020] which attempt to rigorously evaluate the effect of overparameterization level on performance.

2. **Algorithms.** We will train each model using ERM as a baseline and then implement adversarial training with multi-step PGD in the sense of Madry et al. [2017]. If time permits, we will also implement FGSM Goodfellow et al. [2014] and FOSC curriculum learning Wang et al. [2019] for additional comparisons.

3. **Results plots.** Our main results will be plots of (1) ERM adversarial accuracy vs model size (2) multi-step PGD adversarial accuracy vs model size and (3) FOSC criteria vs model size We will use plot (1) as a baseline and plots (2) and (3) to answer our three key questions above.

8 Experiments

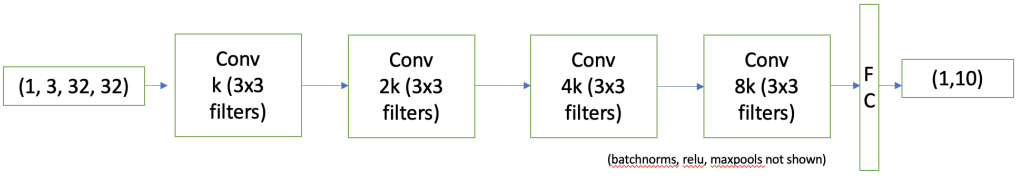


Figure 2: Doubling width CNN, initial width $= k$

The initial widths k used in our experiments are $2^4, 2^5, 2^6, 2^7, 2^8$, epochs=50, no dropout, weight decay $= 10^{-4}$, and we use crop + flip for data augmentation.

8.1 Adversarial accuracy of an ERM trained network with overparameterization

We first study the effects on overparameterization on adversarial accuracy in isolation i.e without adversarial training. The doubling width CNNs are trained normally(ERM), and during test time we plot its performance against a 20-step PGD attack in Figure 3. The adversarial accuracy is slightly better than random prediction(e.g. for intial width $> 2^7$). We also observe that overparameterization helps adversarial accuracy. This is in line with [Bubeck and Sellke, 2021] and suggests that overparameterization is necessary for achieving adversarial robustness in the sense of a constant Lipschitz factor.

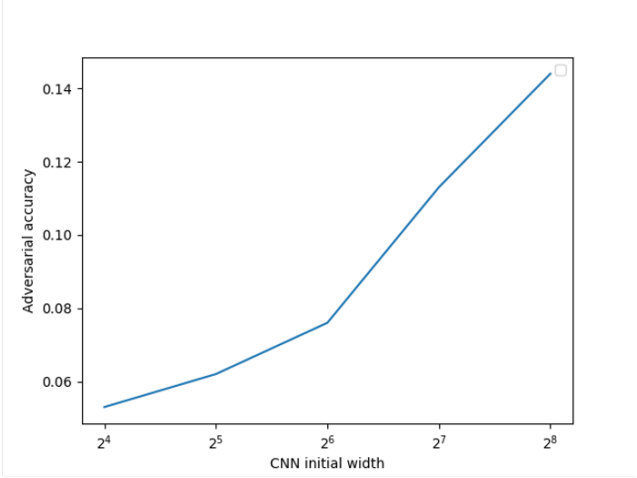


Figure 3: Adversarial accuraccy vs model width (ERM)

8.2 Adversarial accuracy of an PGD trained network with overparameterization

We now study how the extent of adversarial training and overparameterization affect adversarial accuracy. We performed adversarial training with increasing PGD step sizes $(0$ *i.e. ERM*$, 5, 10, 20)$ on each of the doubling width CNNs. During test time we plot the performance against a 20-step PGD attack in figure 4. From the plot its evident that increasing PGD steps during training, increasing overparameterization both improve adversarial accuracy. An interesting observation is that training with 5 PGD steps on CNN of initial width 2^8 has equivalent adversarial accuracy to training with 10 PGD steps on a CNN of intial width 2^4, i.e overparameterizing models may reduce the number of PGD steps while having similar adversarial accuracy.

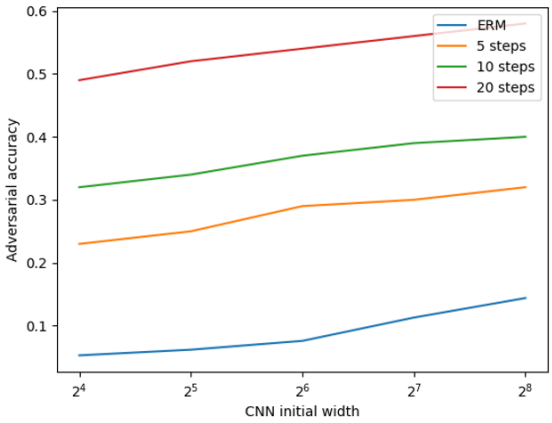


Figure 4: Adversarial accuraccy vs model width (when adversarially trained)

8.3 FOSC vs model size

To study how the efficiency of adversarial training changes with model size, we plot the FOSC criteria reached when we adversarially attack models of different sizes. Specifically, we plot the percent increase in the FOSC score from the smallest model, as we increase model size (we essentially normalize the FOSC score based on the smallest model). We plot this for models adversarially trained with 10 steps of PGD (in this case we attack with 10 steps of PGD) and models adversarially trained with 20 steps of PGD (in this case we attack with 20 steps of PGD). The results can be seen in Figure 5.

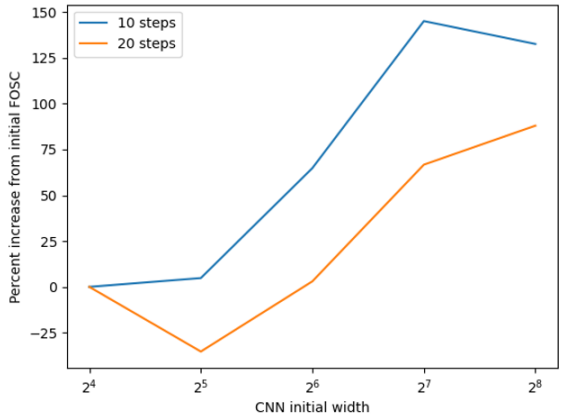


Figure 5: FOSC rise vs model size

From Figure 5, we can see that the FOSC criteria reached drastically increases as we increase model size. This indicates that the adversarial attack achieved is weaker as we increase model size. So while the earlier figures indicated that overparametrized models are more robust, Fig 5 implies that they require more PGD steps for adversarial training. Therefore, optimizing large models with PGD will be less efficient than for smaller models. Interestingly, this finding contradicts the theory that a single gradient step can find an adversarial attack in overparametrized models Bubeck et al. [2021].

8.4 Convergence of FOSC versus Model Size

To further study the effect of model size on the efficiency of adversarial training, we investigate how many steps are needed to solve the inner maximization problem as we vary the model size. For this experiment, we count the number of steps needed until the gradient ascent in the inner maximization converges. We say that the inner maximization has converged once the FOSC score has stopped decreasing. We use models that have been trained with a 10-step PGD attack. The results are shown in Figure 6.

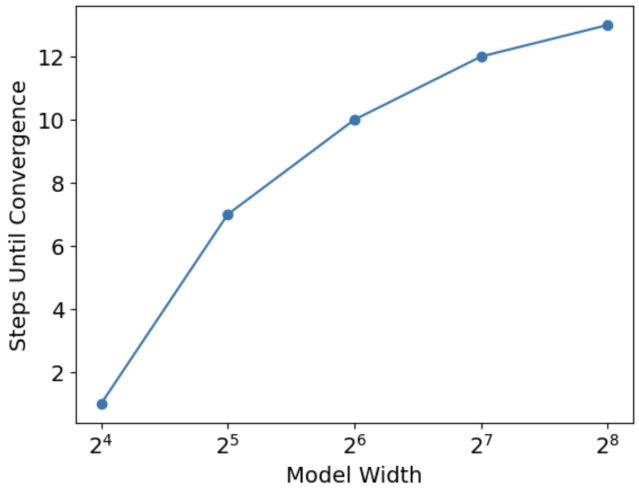


Figure 6: Gradient steps needed in the inner maximization versus model size.

From Figure 6, we can see that for small models, even if they are adversarially trained), the FOSC criteria quickly converges, sometimes requiring only one step to generate a powerful attack. As the model width increases, we notice that more gradient steps are needed to reach convergence, indicating that adversarially training on larger models will take longer than adversarial training on smaller

models. This supports the findings in section 8.3, that adversarial training on overparametrized models may be less efficient.

9 Conclusion

In this report we studied efficient adversarial training of neural networks empirically. In particular, we studied the effect of overparametrization on the robustness and efficiency of adversarial training, using tools such as the FOSC criteria. Our main finding is that increasing the model size improves the adversarial robustness of neural networks, but reduces the efficiency of neural network training. Crucially, our experiments on practical neural networks support some recent theoretical works and contradict others, indicating that there is a large gap between the settings considered in these papers and ones used for computer vision applications. As future work, we would like to theoretically investigate the behavior of adversarial training robustness and efficiency in infinite-width regime, and see if our experimental findings hold true.

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